

CSE321 Assignment 01

Research and Compare the multithreading libraries for the following programming languages and environments:

- **Python threads**
- **POSIX pthreads**
- **Swift threads**

Report Requirements:

Write a comprehensive analysis focusing on the following key factors:

Library Implementation:

- **Describe** how each threading library is structured and its primary features.
- **Discuss** whether it supports preemptive or cooperative multitasking.

OS Integration:

- **Explain** how each library interacts with the underlying operating system.
- **Identify** whether the library is platform-dependent or cross-platform.
- **Discuss** any performance optimizations provided by the OS for these threading models.

Mapping Between User-Level and Kernel-Level Threads:

- **Determine** whether the threading model is based on user-level threads, kernel-level threads, or a hybrid approach.
- **Analyze** how threads are scheduled and managed by the OS.
- **Compare** the efficiency and overhead of each threading approach in different scenarios.

Submission guidelines: Can be found in the submission form given below.

Submission link:

<https://forms.gle/dEX3pGwf72Ls5NjR6>

Deadline: 24/7/2026, 11:00 PM

(Submissions outside the Google Form will be automatically rejected.)